

Martin Allen

git: github.com/martyall

site: martyall.github.io

email: martyall@protonmail.com

EXPERIENCE

o1-Labs, Protocol Engineer, Remote *June 2024 - Present*

- Formalized Kimchi proof system in Lean with proof of soundness
- Implemented rust based Pickles verifier, deployed as SP1 zkvm guest program
- Maintained, refactored and improved the OCaml Mina node internals.
- Core author of Ensue DB, a database layer for verifiable shared agent memory

ℓ-adic Technologies, Core Dev, Remote *July 2023 - Present*

- Implemented an embedded R1CS based circuit DSL in the Haskell programming language
- Built novel arkworks and circom/snarkjs integrations to enable Groth16 proof verification on Ethereum
- Implemented an optimized, arkworks-compatible version of the Bulletproofs protocols.

August Health, Senior Software Engineer, Remote — *Jan 2022 - March 2023*

- Introduced modern functional programming patterns and libraries into a large Scala codebase, helped promote and explain core concepts to our team
- Instituted application metrics and monitoring to identify and resolve critical performance bugs
- Developed various API features and enhancements based on customer requirements

Tweag, Software Engineer (Consulting), Remote — *July 2021 - Jan 2022*

- Made core contributions to the Djed stablecoin application deployed to Cardano, including exhaustive state machine tests and property-based tests before the launch
- Authored the Nix derivations for building, packaging, and deploying the Djed smart contracts

FOAM, Senior Engineer, Founding Member, NYC — *Sept 2017 - Jan 2021*

- Led a team of engineers in both protocol R&D and end-to-end application development
- Architected and implemented novel backend services to support Ethereum applications, including REST APIs, notification systems, and novel indexers/caching for smart contract data.
- Implemented the numerical analysis and statistical filtering backing FOAM's custom geolocation hardware stack
- Developed Ethereum scaling solutions, including Cosmos integrations for the FOAM stack

BlockApps, Senior Platform Engineer, NYC — *Jan 2017 - Sept 2017*

- Implemented and improved core features of BlockApps' Haskell Ethereum client, including the EVM
- Extended the client's proprietary cache layers and relational database plug-ins supporting the developer platform
- Designed and implemented novel Ethereum solutions for private enterprises, specifically in the areas of supply chain management and identity and access management systems.

VFILES, Full Stack Engineer, NYC — *Jan 2016 - Jan 2017*

- Developed and enhanced the REST API and notification systems (written in Haskell) using Postgres and Elasticsearch. Implemented various features on a large, reactive frontend in TypeScript.

EDUCATION

PhD candidate, Mathematics, Central European University, Budapest, Hungary — *2014 - 2016*

- Thesis: *Zeta functions of Normal Surface Singularities*

MSc, Mathematics, University of Paris-Saclay, Orsay, France — *2012 - 2014*

- Thesis: *The Riemann Hypothesis for Hypersurfaces over Finite Fields*

BSc, Mathematics, University of Washington, Seattle, Washington — *2006 - 2010*

OPEN SOURCE PROJECTS: <https://martyall.github.io/open-source-work.html>

LANGUAGES: Haskell, Rust, Scala, PureScript, TypeScript, Python, OCaml, Nix, Dhall, Solidity, Plutus

TECHNOLOGIES: Linux, Nix, Postgres, RocksDB, Elasticsearch, Redis, ZeroMQ, Docker, Terraform